Chapter 3： The second law of thermodynamics

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Summary of Chapter 2

1. The establishment of the **relationship** between the work done on or by a system and the thermal energy entering or leaving the system is facilitated by the introduction of the thermodynamic function U, the internal energy.

2. U is a function of state, and thus, the difference between the values of U in two states depends only on the states and is independent of the path taken by the system in moving between the states.

3. The relationship between the internal energy change, the work done, and the thermal energy absorbed per mole by a system of fixed composition in moving from one state to another is given as $\Delta U = q - w$, or, for an increment of this process, $dU = \delta q - \delta w$. This relationship summarizes the First Law of Thermodynamics.

Summary of Chapter 2

4. The internal energy of an isolated system (δq=0 and δw=0) is constant.

5. The integrals of **δq and δw** can only be obtained if the path taken by the system in moving from one state to another is known. Process paths which are convenient for consideration include

a. Constant-volume processes (isochoric) in which $\int \delta w = \int PdV = 0$, if only PdV work is possible.

b. Constant-pressure processes (isobaric) in which $\int \delta w = P\int dV = P\Delta V$ if only PdV work is possible.

c. Constant-temperature (isothermal) processes.

d. *Adiabatic* processes in which q=0. For adiabatic processes, the work needed to take the system from $U=U_1$ to $U=U_2$ is independent of the path.

Summary of Chapter 2

6. For a **constant-volume process** in a simple system, w=0 and $\Delta U = q_V$.

The definition of the constant-volume molar heat capacity as

$$c_V = \left(\frac{\delta q}{dT}\right)_V = \left(\frac{\partial U}{\partial T}\right)_V$$

(which is an experimentally measurable quantity) facilitates the determination of the change in U resulting from a constant-volume process, since $\Delta U = \int_{T_1}^{T_2} c_V \, dT$

7. Consideration of **constant-pressure processes** is facilitated by the introduction of the thermodynamic function H; the enthalpy for 1 mole is defined as $H \equiv U + PV$. Since the expression for H contains only functions of state, H is also a function of state, and thus, the difference between the values of H in two states depends only on the states and is independent of the path taken by the system in moving between them.

Summary of Chapter 2

8. For a constant-pressure process, $\Delta H = \Delta U + P\Delta V = (q_p - P\Delta V) + P\Delta V = q_p$.

The definition of the constant-pressure molar heat capacity as

$$c_P = \left(\frac{\delta q}{dT}\right)_P = \left(\frac{\partial H}{\partial T}\right)_P$$

(which is an experimentally measurable quantity) facilitates the determination of the change in *H* as the result of a constant-pressure process, since $\Delta H = \int_{T_1}^{T_2} c_P \, dT$

9. For an ideal gas, the internal energy *U* is a function only of temperature.

10. $c_p - c_v = R$ for an ideal gas.

Summary of Chapter 2

11. The process path of an ideal gas undergoing a **reversible adiabatic** change of state is described by PV^{γ}=constant, where $\gamma=c_P/c_V$. During an adiabatic expansion, since q=0, the decrease in the internal energy of the system equals the work done by the system.

12. Since the internal energy of an ideal gas is a function only of temperature, the internal energy of an ideal gas remains constant during an **isothermal** change of state. Thus, the thermal energy which enters or leaves the gas as a result of the isothermal process equals the work done by or on the gas, with both quantities being given by

$$w = q = RT \ln \frac{V_2}{V_1} = RT \ln \frac{P_1}{P_2}$$

Summary of Chapter 2

13. Only the differences in the values of U and H between two states—that is, the values of ΔU and ΔH — can be measured. The absolute values of U and H in any given state cannot be determined.

14. **Other work** terms include the work done on a material by the application of an external magnetic field or by an electric field. Also, the creation of new surface must be done by doing work on the material. A generalized First Law can be written as.

$$dU = \delta q - \sum \delta w_i$$

Each of the terms δw_i is the work done by the system.

Summary of Chapter 2

Adiabatic process; Calorie; Conservation of energy

Cyclic process; Electric work

Enthalpy of melting/freezing; Enthalpy, *H*

First Law of Thermodynamics

Heat (thermal energy); Heat capacity

Hess’ law of constant heat summation

Internal energy, *U;* Isobaric process

Isochoric process; Isothermal process

Kinetic energy/potential energy; Magnetic work

Mechanical equivalent of heat; Process

Reversible processes; Specific/molar heat capacity

Surface energy/work; Thermodynamic state variable (function); Work

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3.1. Introduction

- From the first law, when a system undergoes a change of state, the consequent change in the internal energy of the system, which is dependent only on the initial and final states, is equal to the algebraic sum of the heat and work effects.
- The question now arises: What magnitudes may the heat and work effects have, and what criteria govern these magnitudes?
- Two obvious cases occur, namely, the extreme cases in which either w=0 or q=0, in which cases, respectively, $q=\Delta U$ and $w=-\Delta U$.
- If $q\neq0$ and $w\neq0$, is there a definite amount of work which the system can do during its change of state?

The answers to these questions require an examination of the **nature** of processes.

3.1. Introduction

- This examination, which will be made here, identifies two classes of processes (reversible and irreversible processes) and introduces a state function called the entropy, *S*. The concept of entropy will be introduced from two different starting points.

- Entropy will be introduced and discussed as a result of a need for quantification of the degree of irreversibility of a process;

- It will be seen that, as a result of an examination of the properties of reversibly operated heat engines, there naturally develops a quantity which has all the properties of a state function.

3.2 Spontaneous or natural processes

- Left to itself, a system will do one of two things.

- If the system is initially in equilibrium with its surroundings, then, left to itself, it will remain in this, its equilibrium, state.

- If the initial state is not the equilibrium state, the system will spontaneously move toward its equilibrium state.

- The equilibrium state is a state of rest, and thus, once at equilibrium, a system will only move away from equilibrium if it is acted on by some external agency. Even then, the combined system, comprising the original system and the external agency, is simply moving toward the equilibrium state of the combined system.

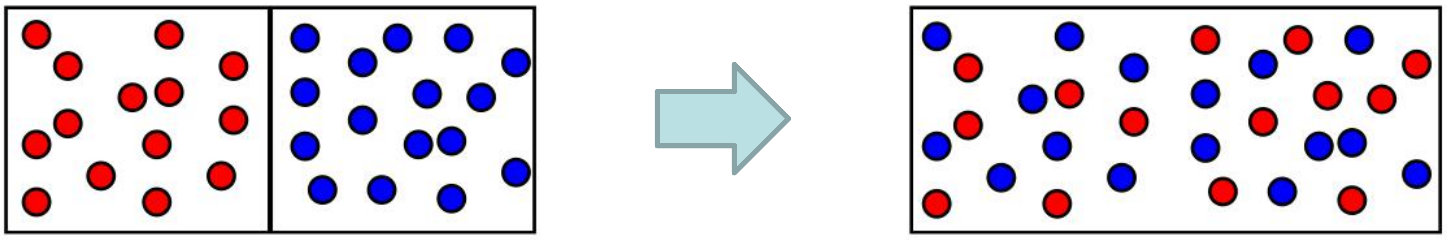
3.2 Spontaneous or natural processes

- A process which involves the spontaneous movement of a system from a non-equilibrium state to an equilibrium state is called **a natural or spontaneous process**.
- As such a process cannot be reversed without the application of an external agency (a process which would leave a permanent change in the external agency), such a process is said to be irreversible.
- The terms natural, spontaneous, and irreversible are synonymous in this context.

3.2 Spontaneous or natural processes

- The mixing of gases and the flow of heat down a temperature gradient are common examples of natural processes.

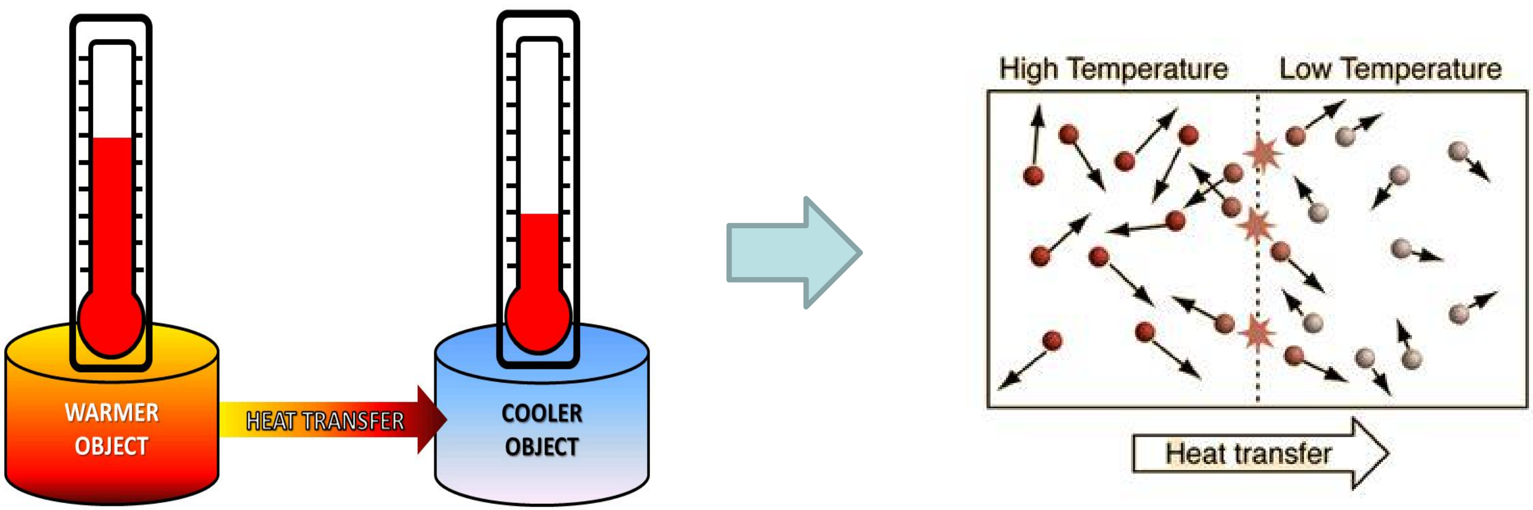
- If the initial state of a system consisting of two gases, A and B, is that in which gas A is contained in one vessel and gas B is contained in separate vessel, then, when the vessels are connected to one another, the system spontaneously moves to the equilibrium state in which the two gases are completely mixed, i.e., the composition of the gas mixture is uniform throughout the volume which the gas occupies.



3.2 Spontaneous or natural processes

◆ The mixing of gases and the flow of heat down a temperature gradient are common examples of natural processes.

- If the initial state of a two-body system is that in which one body is at one temperature and the other body is at another temperature, then, when the bodies are placed in thermal contact with one another, the spontaneous process which occurs is the flow of heat from the hotter to the colder body, and the equilibrium state is reached when both bodies attain a common uniform temperature.



3.2 Spontaneous or natural processes

- In both of these examples the reverse process (unmixing of the gases and the flow of heat up a temperature gradient) will never occur spontaneously,
- In both examples the simplicity of the system, along with common experience, allows the equilibrium states to be predicted without any knowledge of the criteria for equilibrium.

◆ In less simple systems, the equilibrium state cannot be predicted from common experience, and the criteria governing equilibrium must be established before calculation of the equilibrium state can be made.

3.2 Spontaneous or natural processes

- Determination of the equilibrium state is of prime importance in thermodynamics, as knowledge of this state for any materials reaction system will allow determination to be made of the direction in which any reaction will proceed from any starting or initial state.

- For example, knowledge of the equilibrium state of a chemical reaction system such as

$$A + B = C + D$$

will afford knowledge of whether, from any initial state—which would be some mixture of A, B, C, and D—the reaction will proceed from right to left or from left to right, and, in either case, to what extent before equilibrium is reached.

3.2 Spontaneous or natural processes

- If a system undergoes a spontaneous process involving the performance of work and the production of heat, then, as the process continues, during which time the system is being brought nearer and nearer to its equilibrium state, the capacity of the system for **further spontaneous change** decreases. Once equilibrium is reached, the capacity of the system for **doing further work** is exhausted.

- As a result of the spontaneous process, the system has become **degraded**, in that energy, which was available for doing useful work, has been converted to thermal energy (or heat), in which form it is not available for external purposes.

The dissipation of energy

3.3 Entropy and the quantification of irreversibility

Two distinct types of spontaneous process are:

- The conversion of work to heat (i.e., the degradation of mechanical energy to thermal energy)
- The flow of heat down a temperature gradient.

◆ If it is considered that an irreversible process is one in which the energy of the system undergoing the process is degraded, then the possibility that the extent of degradation can differ from one process to another suggests that a *quantitative* measure of the extent of degradation, or degree of irreversibility, exists.

3.3 Entropy and the quantification of irreversibility

- ✓ The existence of processes which exhibit **differing degrees of irreversibility** can be illustrated as follows.

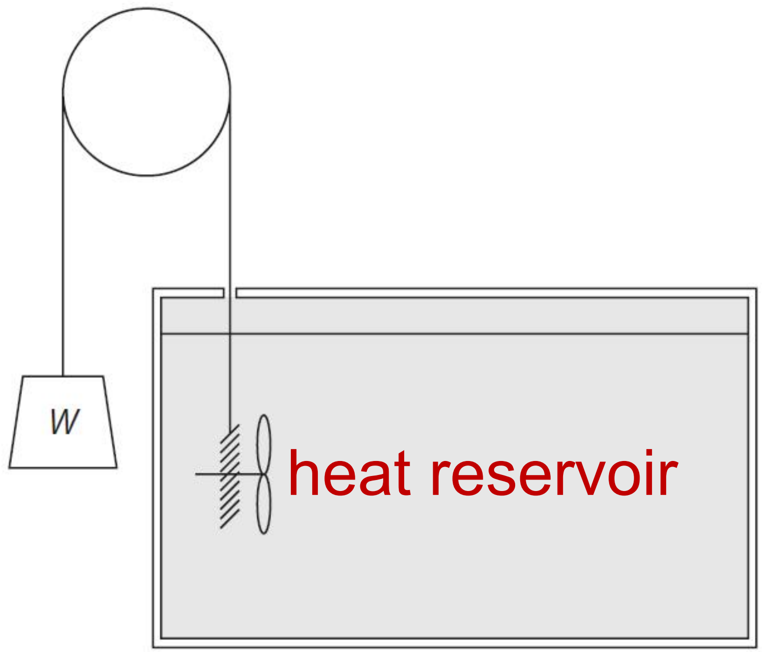


Figure 3.1 A weight–pulley–heat reservoir arrangement in which the work done by the falling weight is degraded to thermal energy, which appears in the heat reservoir.

3.3 Entropy and the quantification of irreversibility

Lewis and Randall considered the following three processes:

1. The heat reservoir in the weight-heat reservoir system is at the temperature T_2. The weight is allowed to fall, performing work, *w*, and the heat produced, *q*, enters the heat reservoir.
2. The heat reservoir at the temperature T_2 is placed in thermal contact with a heat reservoir at a lower temperature T_1, and the same heat *q* is allowed to flow from the reservoir at T_2 to the reservoir at T_1.
3. The heat reservoir in the weight-heat reservoir system is at the temperature T_1. The weight is allowed to fall, performing work, *w*, and the heat produced, *q* enters the reservoir.

Each of these process is spontaneous and hence irreversible, and degradation occurs in each of them.

3.3 Entropy and the quantification of irreversibility

- As process (3) is the sum of processes (1) and (2), the degradation occurring in process (3) must be greater than the degradation occurring in each of the processes (1) and (2). Thus it can be said that process (3) is more irreversible than either process (1) or process (2).

- Examination of the three processes indicates that both the amount of heat produced, q, and **the temperatures between which this heat flows are important in defining a quantitative scale of irreversibility**.

- In the case of comparison between process (1) and process (3), the quantity q/T_2 is smaller than the quantity q/T_1, being in agreement with the conclusion that process (1) is less irreversible than process (3).

3.3 Entropy and the quantification of irreversibility

- The quantity q/T is thus taken as being a measure of the degree of irreversibility of the process, and the value of q/T is called the increase in entropy, S, occurring as a result of the process.

✓ When the weight-heat reservoir system undergoes a spontaneous process which causes the absorption of heat q at the constant temperature T, the entropy produced by the system, ΔS, is given by:

$$\Delta S = \frac{q}{T}$$

The increase in entropy, caused by the process, is thus a measure of the degree of irreversibility of the process.

3.4 Reversible Processes

- ✓ As the degree of irreversibility of a process is variable, it should be possible for the process to be conducted in such a manner that the degree of irreversibility is minimized.

- The ultimate of this minimization is a process in which the degree of irreversibility is zero and in which no degradation occurs.

- This limit, toward which the behavior of real systems can be made to approach, is the reversible process. If a process is reversible, then the concept of spontaneity is no longer applicable.

- Spontaneity occurred as a result of the system moving, of its own accord, from a non-equilibrium state to an equilibrium state.

3.4 Reversible Processes

- **A reversible process** is the one during which the system is never away from equilibrium, and a reversible process which takes the system from the state A to the state B is one in which the process path passes through a continuum of equilibrium states. Such a path is, of course, imaginary.
- ***a quasi-static process*** is one which proceeds under the influence of an infinitesimally small driving force such that, during the process, the system is never more than an infinitesimal distance from equilibrium.

- If, at any point along the path, the external influence is removed, then the process ceases.
- If the direction of the external influence is reversed, then the direction of the process is reversed.

3.5 Illustration of irreversible and reversible processes

- Consider the reversible isothermal expansion of 1 mole of a monatomic ideal gas from the state (V_A ,T) to the state (V_B ,T), where $V_B>V_A$. The gas is placed in thermal contact with a heat reservoir at the temperature T, and by slowly reducing the weight on the piston by removing one grain of sand at a time, the pressure exerted by the gas is only infinitesimally greater than the instantaneous pressure exerted by the piston on the gas.

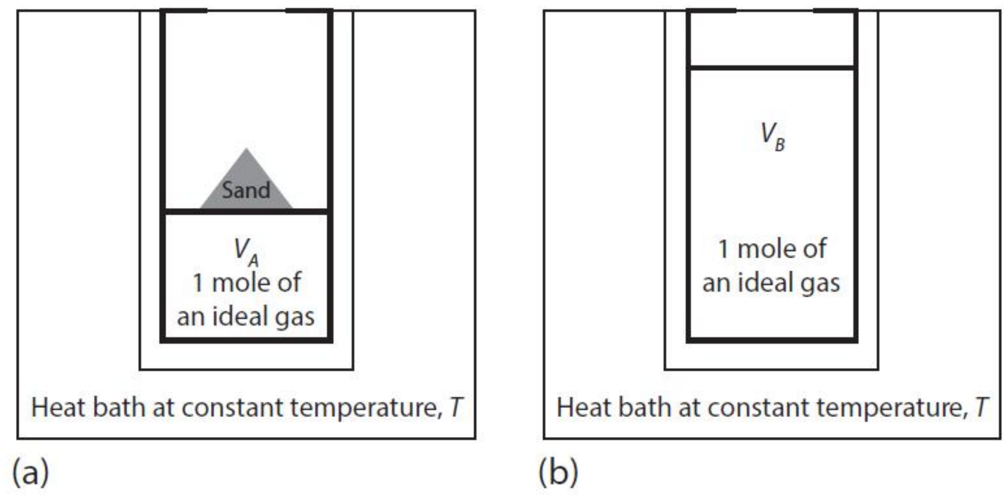


Figure 3.2 (a) One mole of a confined ideal gas being held at V_A by a mass of sand. (b) The mass of sand is removed and the new volume if the gas is V_B.

3.5 Illustration of irreversible and reversible processes

- The state of the gas thus lies, at all times, on a section at the constant temperature T of the V-P-T surface, and hence, the gas passes through a continuum of equilibrium states in going from the state (V_A ,T) to the state (V_B ,T). Since the gas is never out of equilibrium, the process is reversible.

$$\Delta U = q - w$$

- The internal energy of an ideal gas depends only on its temperature. Hence, ΔU=0, and thus, $q = w$; that is, the work done by the expanding gas on the piston equals the thermal energy transferred from the constant-temperature heat reservoir into the gas.

$$w_{rev} = \int_{V_A}^{V_B} P dV = \int_{V_A}^{V_B} \frac{RT}{V} dV = RT \ln \frac{V_B}{V_A}$$

3.5 Illustration of irreversible and reversible processes

- Since $V_B > V_A$, w_{rev} is a positive quantity, in accordance with the fact that work is done *by* the gas. The transfer of thermal energy from the reservoir to the gas (at constant internal energy) causes a change in the entropy of the gas: $$\Delta S_{\text{gas}} = S_B - S_A = \frac{q_{rev}}{T} = \frac{w_{rev}}{T} = R\ln\frac{V_B}{V_A} > 0$$

The change in the entropy of the reservoir is given by:

$$\Delta S_{\text{heat reservoir}} = -\frac{q}{T} = -\Delta g_{gas} = R\ln\frac{V_A}{V_B} < 0$$

The total change in the entropy can be written as:

$$\Delta S_{\text{total}} = \Delta S_{\text{gas}} + \Delta S_{\text{heat reservoir}} = 0$$

For the reversible process, the total change in entropy of the universe is zero: no entropy has been produced

3.5 Illustration of irreversible and reversible processes

- Consider the free expansion (i.e., expansion against zero atmospheric pressure) of 1 mole of an ideal gas from V_A to V_B. The free expansion of an ideal gas is also isothermal. Thus, the final state of the free expansion is the same as the state of the isothermal reversible expansion process (i.e., both states have the same V and T).

- In this case, no work is done by the gas against the piston, since the weight on the piston is rapidly removed. Thus, $\Delta U=0$ and $w=0$, which by the First Law means that $q=0$ as well. Since entropy is a state function, the change of entropy of the gas for the free expansion must be the same as that for the isothermal expansion.

$$\Delta S_{\mathrm{gas}} = R \ln \frac{V_B}{V_A}$$

3.5 Illustration of irreversible and reversible processes

- Since no thermal energy leaves the heat reservoir, $\Delta S_{reservoir}=0$. The total change in the entropy (gas plus heat reservoir) for the free expansion is

$$\Delta S_{\text{total}} = \Delta S_{\text{gas}} + \Delta S_{\text{heat reservoir}} = \Delta S_{\text{gas}} = R\ln\frac{V_B}{V_A}$$

In the case of free expansion, there is no decrease in entropy of the heat reservoir, since no thermal energy was absorbed by the ideal gas, since it did no work. Thus, for the free expansion process, the entropy of the universe increases ($\Delta S_{total} > 0$).

3 .6 Further differences between reversible and irreversible expansion

- We now look at the processes from the point of view of the thermal energy that was transferred from the heat reservoir to the gas and the work that was performed by the gas.

- For the reversible isothermal expansion case, $q_{rev}=w_{max}$.

✓ This process yields the maximum amount of work that can be performed by the gas during isothermal expansion.

✓ The reversible process also yields the most amount of thermal energy that can be transferred to the gas from the heat reservoir.

✓ The total change in entropy of the universe is zero.

3 .6 Further differences between reversible and irreversible expansion

- ◆ We now look at the processes from the point of view of the thermal energy that was transferred from the heat reservoir to the gas and the work that was performed by the gas.
- For the case of the free expansion of the gas, we have seen that $w=q = 0$.
- ✓ There is no decrease in entropy of the heat reservoir, since no thermal energy was absorbed by the ideal gas, since it did no work.
- ✓ The entropy of the universe increases ($\Delta S_{total} > 0$).

3 .6 Further differences between reversible and irreversible expansion

$$0 \leq w \leq w_{\max} \qquad 0 \leq q \leq q_{rev} \qquad 0 \leq \Delta S_{total} \leq \frac{q_{rev}}{T}$$

- $\Delta S_{\text{total}} = 0$ when the process is reversible.
- $\Delta S_{\text{total}} > 0$ when the process is irreversible.
- The maximum value of ΔS_{total} occurs for the completely irreversible free expansion case.

3 .6 Further differences between reversible and irreversible expansion

- It is important to note that the difference in entropy between the final and initial states of the gas is independent of whether the process is conducted reversibly or irreversibly. In going from state A to state B ,

$$\Delta S = S_B - S_A = \frac{q}{T} + \Delta S_{irr} = \frac{q_{rev}}{T}$$

  - Since the change in entropy can be determined only by the measurement of thermal energy transferred reversibly at the temperature *T*, then entropy changes can be measured only for reversible processes, in which case the measured thermal energy transferred is q_{rev} and $\Delta S_{irr} = 0$.

3 .6 Further differences between reversible and irreversible expansion

◆ Consider further an evaporation process.

- The work done by the system during the evaporation of 1 mole is seen to have its maximum value, $w_{max}=P_{ext}V$, when the process is conducted reversibly.
- Any irreversible evaporation process performs less work, $w=(P_{ext}-\Delta P)V$.

3 .6 Further differences between reversible and irreversible expansion

- From the First Law, the maximum amount of heat, q_{rev}, is transferred from the heat reservoir to the cylinder when the process is carried out reversibly, where $q_{rev}=\Delta U+w_{max}$.
- If the process is carried out irreversibly, then less heat q is transferred from the reservoir to the cylinder, where $q=\Delta U+w$.
- The difference between the work done during the reversible process and that done during the irreversible process, $(w_{max}-w)$, is the mechanical energy which has been degraded to thermal energy (heat) in the cylinder as a result of the irreversible nature of the process.
- This heat produced by degradation, which is given as $(q_{rev}-q)=(w_{max}-w)$, accounts for the fact that less heat is transferred to the cylinder from the reservoir during the irreversible process than is transferred during the reversible process.

3 .6 Further differences between reversible and irreversible expansion

If the evaporation process is conducted reversibly, then heat q_{rev} leaves the heat reservoir and enters the cylinder at the temperature T.

- The change in the entropy of the heat reservoir is given as:

$$\Delta S_{\text{heat reservoir}} = -\frac{q_{rev}}{T}$$

- The change in the entropy of the water and water vapor in the cylinder is:

$$\Delta S_{\text{water+vapor}} = \frac{q_{rev}}{T}$$

$$\Delta S_{total} = \Delta S_{\text{heat reservoir}} + \Delta S_{\text{water+vapor}} = 0$$

This zero change in entropy is due to the fact that the process was carried out reversibly, i.e., no degradation occurred during the process.

3 .6 Further differences between reversible and irreversible expansion

If the evaporation is carried out irreversibly then heat q ($q<q_{rev}$) is transferred from the reservoir to the cylinder.

- The change in the entropy of the heat reservoir is:

$$\Delta S_{\text{heat reservoir}} = -\frac{q}{T}$$

- The total heat appearing in the cylinder equals the heat q transferred from the heat reservoir plus the heat which is produced by degradation of work due to the irreversible nature of the process.

- Thus degraded work, (w_{max}–w), equals (q_{rev}– q).

3 .6 Further differences between reversible and irreversible expansion

- The change in the entropy of the contents of the cylinder is

$$\Delta S_{\text{water+vapor}} = \frac{q}{T} + \frac{q_{rev} - q}{T} = \frac{q_{rev}}{T}$$

- The change in the entropy of the combined system caused by the irreversible nature of the process is

$$\Delta S_{total} = \Delta S_{\text{heat reservoir}} + \Delta S_{\text{water+vapor}} = \frac{q_{rev} - q}{T}$$

As $q_{\text{rev}} > q$, this entropy change is positive, and thus entropy has been produced (or created) as a result of the occurrence of an irreversible process.

3 .6 Further differences between reversible and irreversible expansion

- The entropy produced, $(q_{rev}–q)/T$, is termed $\Delta S_{\text{irreversible}}$ (ΔS_{irr}) and is the measure of the degradation which has occurred as a result of the process

$$\Delta S_{\text{water+vapor}} = \frac{q}{T} + \frac{q_{rev} - q}{T} = \frac{q}{T} + \Delta S_{irr}$$

$$\Delta S = S_B - S_A = \frac{q}{T} + \Delta S_{irr} = \frac{q_{rev}}{T}$$

3.7 Compression of an ideal gas

Consider the reversible isothermal compression of 1 mole of an ideal gas from the state (V_B ,T) to the state (V_A ,T). The gas is placed in thermal contact with a heat reservoir at the temperature T, and, by adding one grain of sand at a time to the top of the piston, the gas is compressed slowly enough that, at all times during its compression, the pressure exerted on the gas is only infinitesimally greater than the instantaneous pressure of the gas, P_{inst}, where $P_{inst} = RT/V_{\mathrm{inst}}$.

- The state of the gas thus lies, at all times, on a section at the constant temperature T of the V-P-T surface, and hence, the gas passes through a continuum of equilibrium states in going from the state (V_B ,T) to the state (V_A ,T).

3.7 Compression of an ideal gas

- Since the gas is never out of equilibrium, the process is reversible and no degradation of energy occurs. Entropy is not *created* during this process. Entropy is transferred from the gas to the heat reservoir, where it is measured as the thermal energy entering divided by the temperature *T*.
- Since the compression is conducted isothermally, ΔU=0; thus, the work done on the gas is equal to the thermal energy withdrawn from the gas; that is,

$$w_{\max} = q_{rev}$$

$$w_{\max} = \int_{V_B}^{V_A} PdV = \int_{V_B}^{V_A} \frac{RT}{V} dV = RT \ln \frac{V_A}{V_B}$$

3.7 Compression of an ideal gas

- Since $V_B > V_A$, w_{max} is a negative quantity, in accordance with the fact that work is done on the gas. The transfer of thermal energy from the gas to the reservoir causes a change in the entropy of the gas:

$$\Delta S_{gas} = \frac{q_{rev}}{T} = \frac{w_{\max}}{T} = RT \ln \frac{V_A}{V_B} < 0$$

- Since there is no change in the total entropy during the reversible compression, the change in the entropy of the reservoir is given by

$$\Delta S_{\text{heat reservoir}} = -\Delta S_{\text{gas}} = R \ln \frac{V_B}{V_A} > 0$$

3.8 The adiabatic expansion of an ideal gas

Consider the reversible adiabatic expansion of 1 mole of an ideal gas from the state (P_A, T_A) to the state (P_B, T_B), where $P_B < P_A$. For the process to be reversible, it must be conducted slowly enough that, at all times, the state of the gas lies on its *V*-*P*-*T* surface.

- This condition, together with the condition that $q=0$ (an adiabatic process), dictates that the process path across the *V*-*P*-*T* surface follows the curve PV^{γ}=constant.
- Since the process is reversible, no degradation of energy occurs, and, since the process is adiabatic, no thermal energy transfer occurs.
- The change in the entropy of the gas is therefore zero.

3.8 The adiabatic expansion of an ideal gas

- All states of an ideal gas lying on a PV^{γ} = constant curve are states of equal entropy (cf. all states of an ideal gas lying on a PV= constant curve are states of equal internal energy). A reversible adiabatic process is thus an ***isentropic*** process.
- During a reversible adiabatic expansion, the work done by the gas, w_{max}, equals the decrease in the internal energy of the gas; that is.

$$\Delta U = -w_{\max}$$

$$w = \int_{V_A}^{V_B} P dV = K \int_{V_A}^{V_B} V^{-\gamma} dV = \frac{PV}{-\gamma + 1} \Big|_{P_A V_A}^{P_B V_B} = \frac{R\left(T_B - T_A\right)}{1 - \gamma}$$

$$= \frac{3}{2} R\left(T_A - T_B\right)$$

3.8 The adiabatic expansion of an ideal gas

- Since the work done by the gas is positive, $T_B < T_A$; that is, the gas cools. The change in the internal energy can be calculated as:

$$\Delta U = \int_{T_A}^{T_B} c_V \, dT = \frac{3}{2} R \left(T_B - T_A \right) = -w$$

The internal energy of the gas decreased by the amount of work that the gas performed.

- The gas has now returned to its original equilibrium state (P_A, T_A). If the pressure exerted on the gas is now suddenly decreased from P_A to P_B , then the state of the gas moves off the *V*-*P*-*T* surface, and, being out of equilibrium, the expansion occurs irreversibly and the degradation of energy occurs.

3.8 The adiabatic expansion of an ideal gas

- Since the gas is contained adiabatically, the thermal energy produced by the energy degradation remains in the gas, and thus, the final temperature of the gas after an irreversible expansion is greater than the temperature T_B of the reversible adiabatic expansion.
- The final state of a gas after an irreversible adiabatic expansion from P_A to P_B differs from the final state after a reversible expansion from the same initial to the same final pressures.
- The irreversible adiabatic expansion does not follow the path PV^γ = constant. The entropy produced in the gas due to the irreversible process is the difference in entropy between the final and initial states.

3.8 The adiabatic expansion of an ideal gas

- For a given decrease in pressure ($P_A \rightarrow P_B$), the more irreversible the process, the more thermal energy produced in the gas by degradation, the higher the final temperature and internal energy, and the greater the increase in entropy.
- During an irreversible adiabatic expansion, the work done by the gas still equals the decrease in the internal energy of the gas (as is required by the First Law), but the decrease in U is less than that in the reversible expansion from P_A to P_B, due to the production of thermal energy in the gas as the result of degradation.

3.9 Summary statements

1. Entropy is a thermodynamic state variable (function).
2. Entropy is not created when a system undergoes a reversible process; entropy is transferred from one part of the system/surroundings to another part.
3. The total entropy of the universe increases when an irreversible process occurs.
4. For all processes, we can write $\Delta S_{system} = q/T + S_{irr}$ and $q \leq q_{rev}$.
5. For all processes, the entropy of the universe increases or stays the same. The total entropy of the universe never decreases.

The End

Discussion

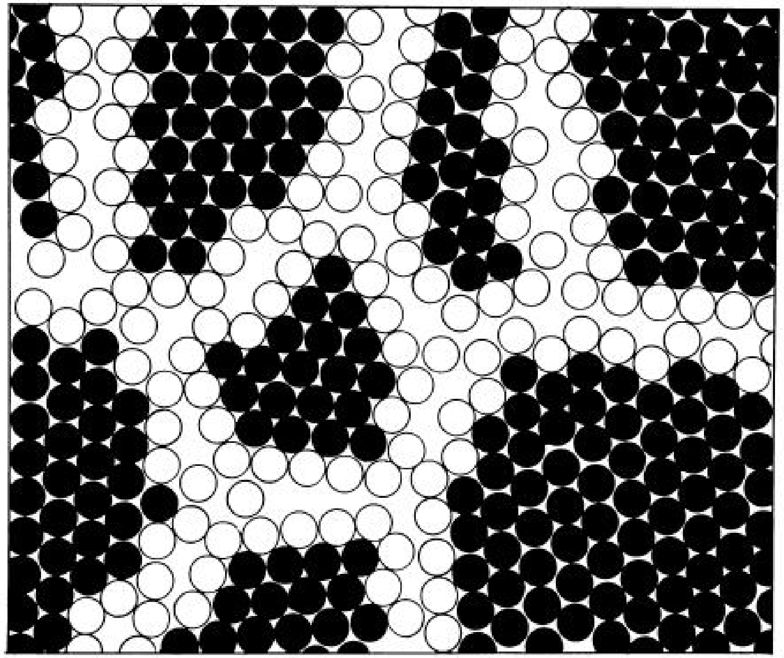


Fig. 2. Two-dimensional model of a nanostructured material. The atoms in the centers of the crystals are indicated in black. The ones in the boundary core regions are represented as open circles [13].

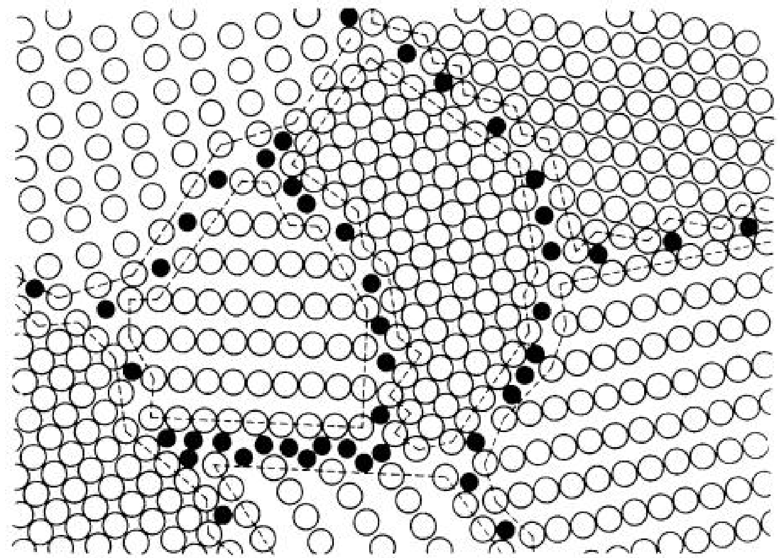


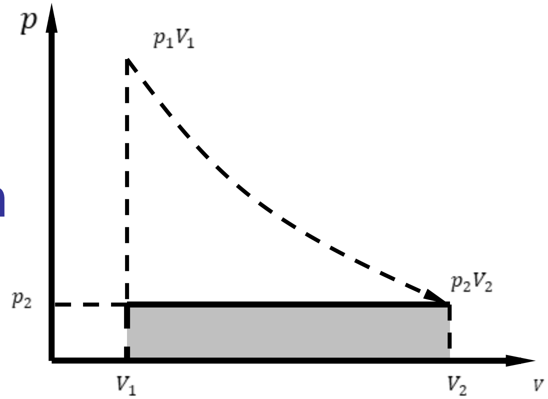
Fig. 4. Schematic model of the structure of nanostructured Cu–Bi and W–Ga alloys. The open circles represent the Cu or W atoms, respectively, forming the nanometer-sized crystals. The black circles are the Bi or Ga atoms, respectively, incorporated in the boundaries at sites of enhanced local free volume. The atomic structure shown was deduced from EXAFS and X-ray diffraction measurements [6].

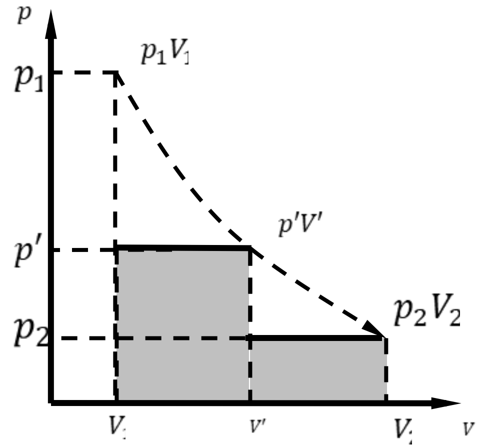
H. Gleiter, Acta mater. 48 (2000) 1-29

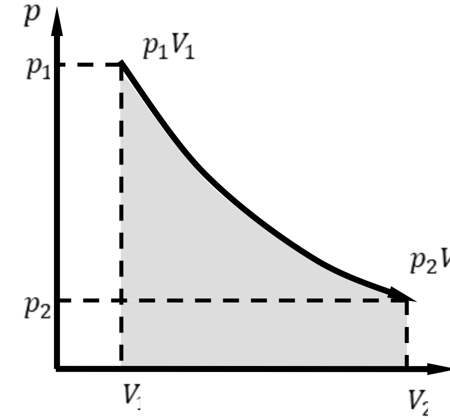
Grain boundary energy

Expansion work under different processes

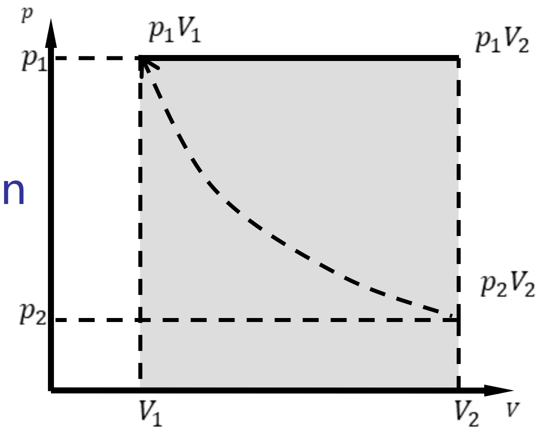
Expansion

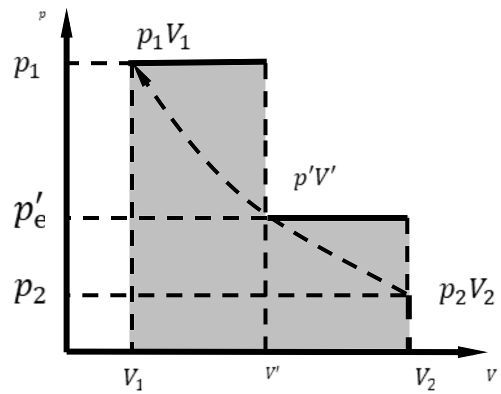


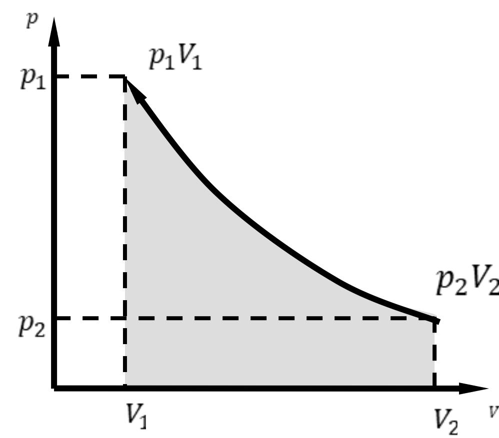




Compression







One time　　Two time　　Reversible

For reversible expansion, the system does the largest work on the surroundings;

For reversible compression, the system does the smallest work on the surroundings.